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Paper Code : PEC- CS801 B/PECIT801B Cryptography & Network Security

UPID : 008257

CS/B.TECH(N)/EVEN/SEM-8/8257/2025-2026/I059

Time Allotted : 3 Hours

Full Marks :70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) In cryptography, the order of letters in a message is rearranged by _____
a) Transpositional ciphers
b) Substitution ciphers
c) Both (a) and (b)
d) Quadratic ciphers
- (ii) Caesar Cipher is an example of
a) Poly-alphabetic Cipher
b) Mono-alphabetic Cipher
c) Multi-alphabetic Cipher
d) Bi-alphabetic Cipher
- (iii) What is data encryption standard (DES)?
a) Block cipher
b) Stream cipher
c) Bit cipher
d) Byte cipher
- (iv) The number of rounds in RC5 can range from 0 to _____
a) 127
b) 63
c) 255
d) 31
- (v) In the DES algorithm the Round Input is 32 bits, which is expanded to 48 bits via _____
a) Scaling of the existing bits
b) Duplication of the existing bits
c) Addition of zeros
d) Addition of ones
- (vi) _____ is the science and art of transforming messages to make them secure and immune to attacks
a) Cryptography
b) Calligraphy
c) Cryptanalysis
d) None of the above
- (vii) In the RSA algorithm, we select 2 random large values 'p' and 'q'. Which of the following is the property of 'p' and 'q'?
a) p and q should be divisible by $\Phi(n)$
b) p and q should be co-prime
c) p and q should be prime
d) p/q should give no remainder
- (viii) A/an _____ algorithm transforms ciphertext to plaintext
a) Encryption
b) Decryption
c) Both (a) and (b)
d) None of the above
- (ix) Which of the following cannot be achieved by the digital signature in symmetric key encryption?
a) Message authentication
b) Data integrity

- c) Non-repudiation
- d) Security of the message
- (X) A packet filter firewall filters at _____
 - a) Physical layer
 - b) Data link layer
 - c) Network layer or Transport layer
 - d) Application layer
- (XI) The full form of SSL is
 - a) Serial Session Layer
 - b) Secure Socket Layer
 - c) Session Secure Layer
 - d) Series Socket Layer
- (XII) When a hash function is used to provide message authentication, the hash function value is referred to as
 - a) Message Field
 - b) Message Digest
 - c) Message Score
 - d) Message Leap

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. Explain confusion and diffusion properties of modern block ciphers. [5]
3. Explain the S-box design of DES algorithm. [5]
4. Compare SSL and TLS. [5]
5. How digital signature is implemented using RSA approach. [5]
6. Explain the working principles of packet filter firewall with proper example. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. (a) Explain the sequence of steps involved in the message generation and reception in Pretty Good Privacy (PGP) with block diagrams. [9]
- (b) List out the security association (SA) parameters in IPSec. [6]
8. (a) Explain the algorithm for generating keys in RSA algorithm. Perform encryption and decryption using RSA Alg. for the following: $P=7$; $q=11$; $e=13$; $M=8$ [9]
- (b) Illustrate man in the middle attack on Diffie Hellman key exchange algorithm. [6]
9. (a) Explain the sequence of steps involved in the message generation and reception in PGP with block diagrams. <https://www.makaut.com> [8]
- (b) List out the benefits of IPSec. [7]
10. (a) Encrypt the word "Semester Result" with the keyword "Examination" using playfair cipher. [8]
- (b) Compare symmetric and asymmetric key cryptography algorithms. [4]
- (c) Convert the Given Text "CRYPTOGRAPHY" into cipher text using Rail fence Technique. [3]
11. Explain the DES and General structure of DES with diagrams. [15]

*** END OF PAPER ***